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MZUZU DIOCESE

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

MATHEMATICS

(100 marks)

Thursday, 23rd March **TIME ALLOWED: 2 HOURS**

(07:30 -09:30) a.m

INSTRUCTIONS

1. This paper has **16** pages. Please check.
2. Answer all 20 questions in this paper
3. The maximum number of marks for each answer is indicated against each question.
4. Write your answers in the spaces provided
5. Calculators may be used and all working should be clearly shown.
6. Please make sure that you write your examination number and class on top of all pages in this paper before you start answering questions.

Questions Answered	Tick if Answered	Do not write in this column	
1			
2			
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Total			

Turn over

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1. Factorise completely $3n^2 - 12n + 15(n - 4)$ **(4 marks)**

2. Given that $g(x) = 3x - 1$. If the image is the set of$\{2, 5, 8\}$, find the set of domains. **(4 marks)**

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3. Simplify by rationalise the denominator: $\sqrt{\frac{2}{5}} + \sqrt{40} + \sqrt{10}$ **(4 marks)**

4. Given that $P = A(1 + \frac{r}{100})^n$, make n subject of the formula. **(4 marks)**

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5. Given that the universal set μ=$\{11, 12, 13, 14, 15, 16, 17, 18, 19\}$.

Set B = $\{numbers\ greator\ than\ 16\}$, Set C = $\{multiples\ of\ 3\}$. Find the element of **set**$(B \cup C)'$. **(4 marks)**

6. Given that $\begin{pmatrix} 1-y & 3 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} 6 \\ x \end{pmatrix} = \begin{pmatrix} -6 \\ x \end{pmatrix}$, find the values of *x* and *y*. **(5 marks)**

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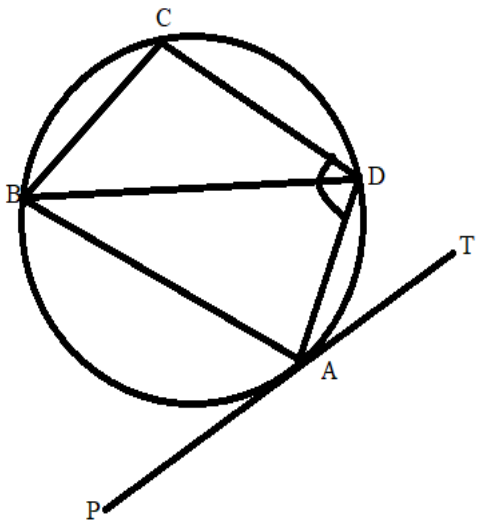
7. **Figure 1** shows circle ABCD and TAP tangent at A. Angle TAD=35° and angle ADC=104°.

Figure 1



Find angle CBD. **(4 marks)**

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8. Solve the equation $\frac{x}{x+3} - \frac{1}{x+1} = \frac{1}{15}$. **(7 marks)**

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9. Form the equation of the line passing through (−4, −5) perpendicular to $y = \frac{-2}{3}x + 1$. **(4 marks)**

10. Given that $\underline{a}\begin{pmatrix}3\\1\end{pmatrix}$ and $\underline{b}\begin{pmatrix}-2\\1\end{pmatrix}$, Find $|\underline{a}| + |\underline{b}|$. **(4 marks)**

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11. A quantity x proportional as the square of z and inversely as y. Given that $x = 2$ when $y = 2$ and $z = 1$, find y when $x = 5$ and $z = -2$. **(5 marks)**

12. The third term of a geometric progression is 360 and the sixth term is 9720. Find the first term of the progression. (**5 marks)**

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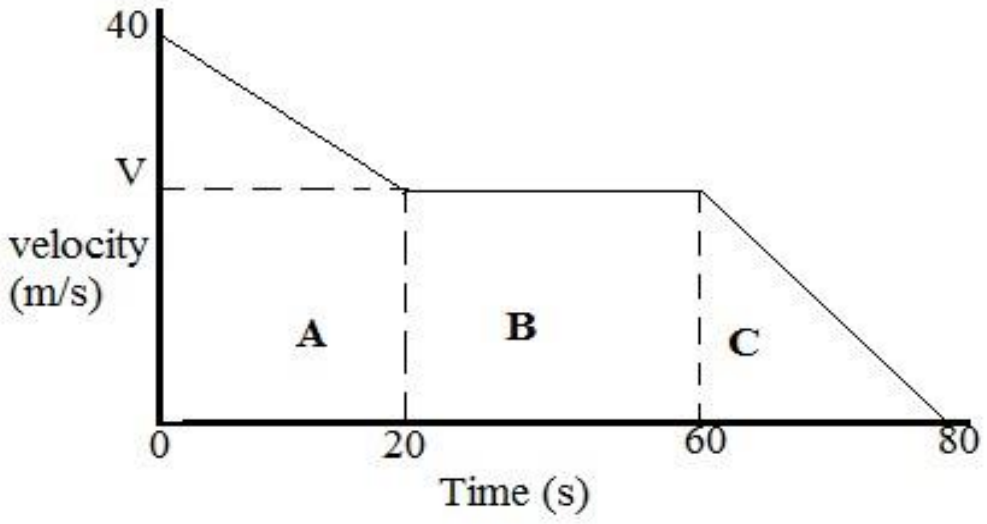
13. Figure 2 shows a speed-time graph of a car. The car decelerates uniformly from 40 m/s to V m/s in 20 seconds. It then maintains this speed for next 40 seconds after which it decelerates uniformly for 20 seconds and stops.



Given that the total distance travelled is 1600m calculate the deceleration during the last 20 seconds. **(6 marks)**

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14. The data below shows marks scored in mathematics test

65, 69, 55, 63, 41, 59, 40, 29, 89, 53, 78, 43, 57, 61, 75,

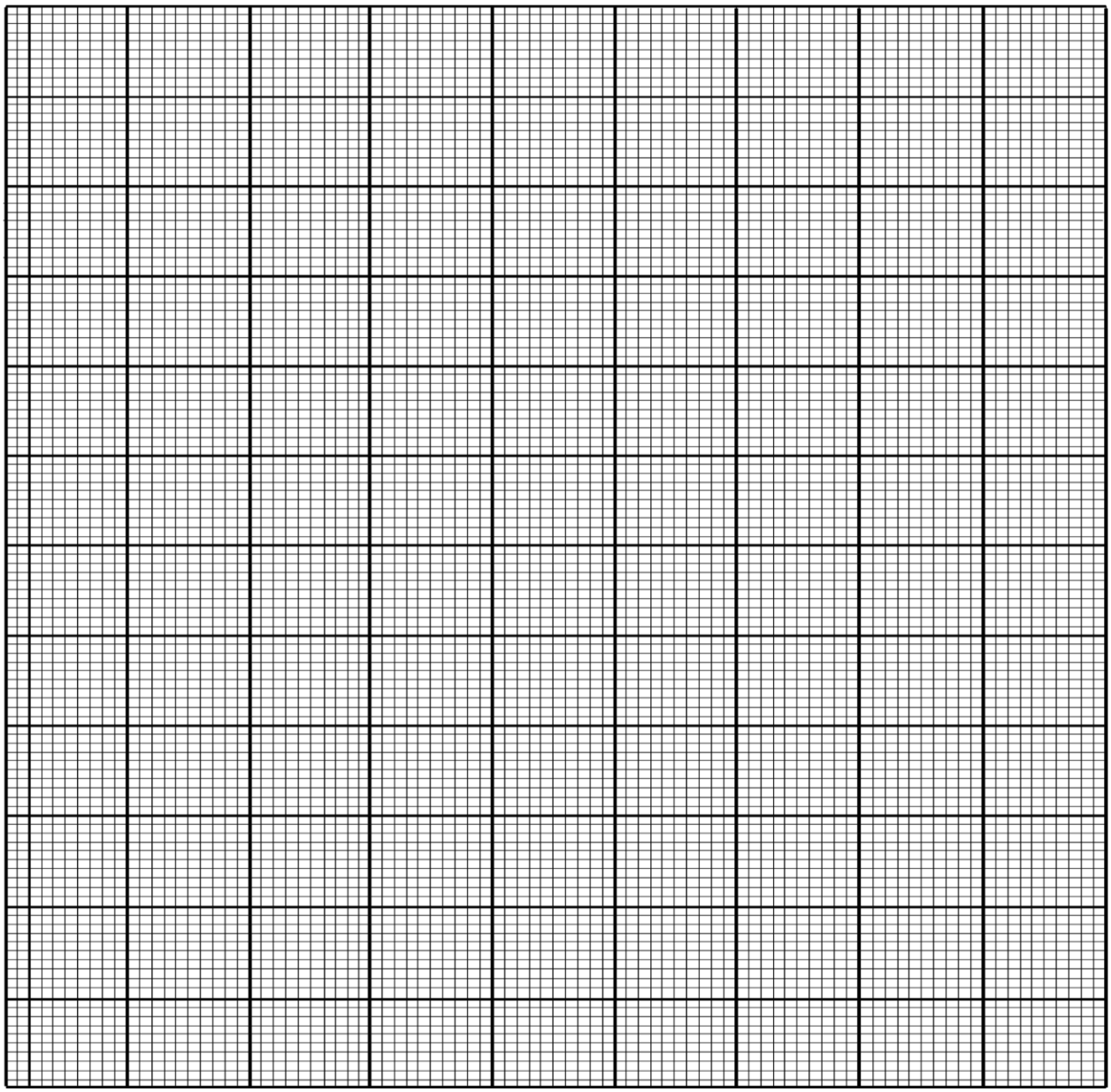
49,83,20, 38, 30, 79, 90, 40, 69, 48, 36, 68, 49, 46, 56, 50, 84,

35, 62.

a. Use the class interval of $20 - 39, 40 - 59,$ ... construct frequency table.

(3 marks)

b. Use frequency table draw frequency polygon to present the information above



(5marks)

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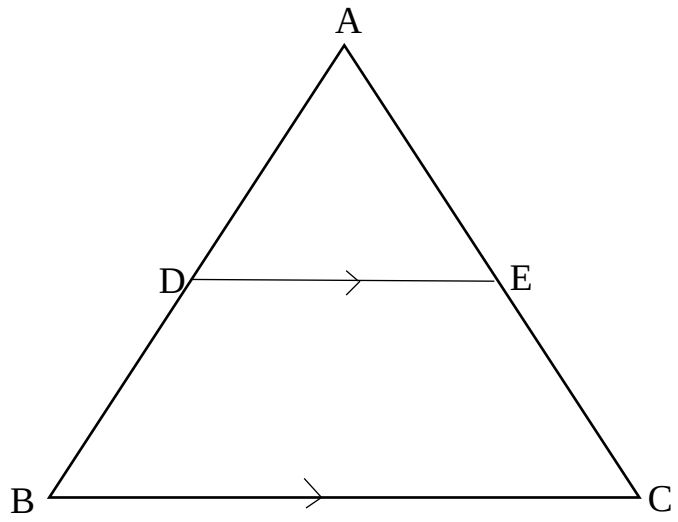
15. Figure 3 below shows triangles ADE and ABC in which AD = 3cm, AB = 5cm and DE is parallel to BC.



If the area of the triangle ADE = $6cm^2$, find the area of the trapezium DECB **(5 marks)**

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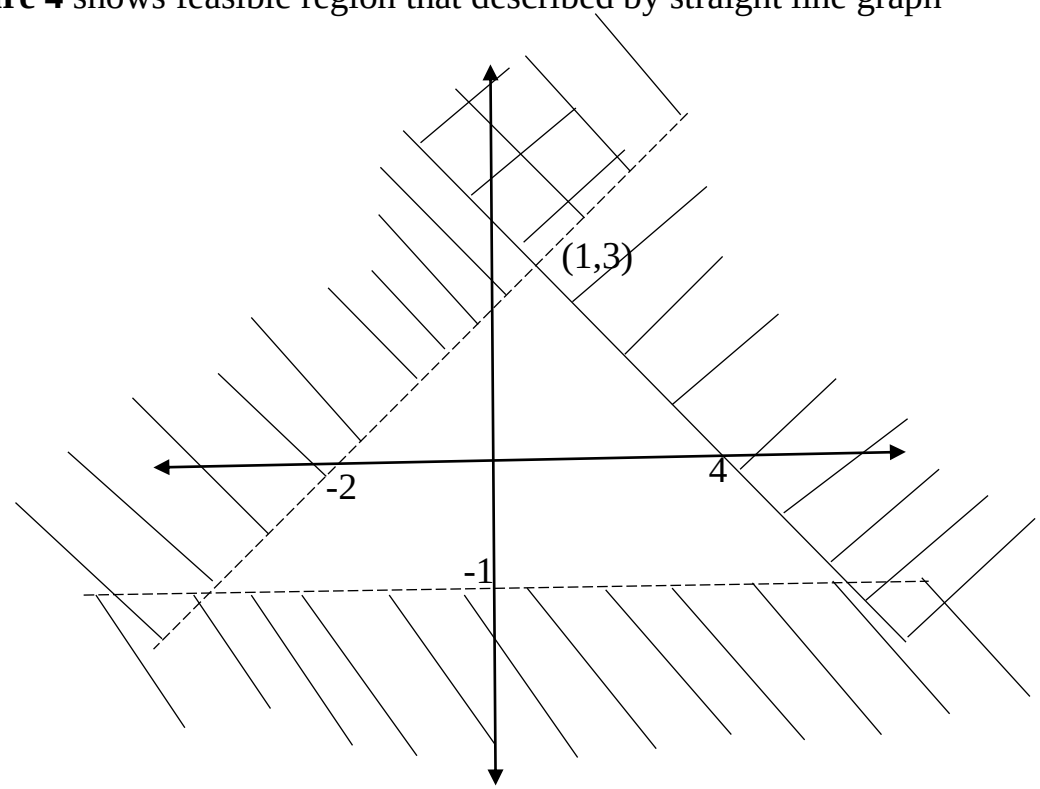
16. Given that $(x+3)(x+1)^2 \equiv Ax^3 + Bx^2 + Cx + D$, find the value of C

(5 marks)

17. Figure 4 shows feasible region that described by straight line graph



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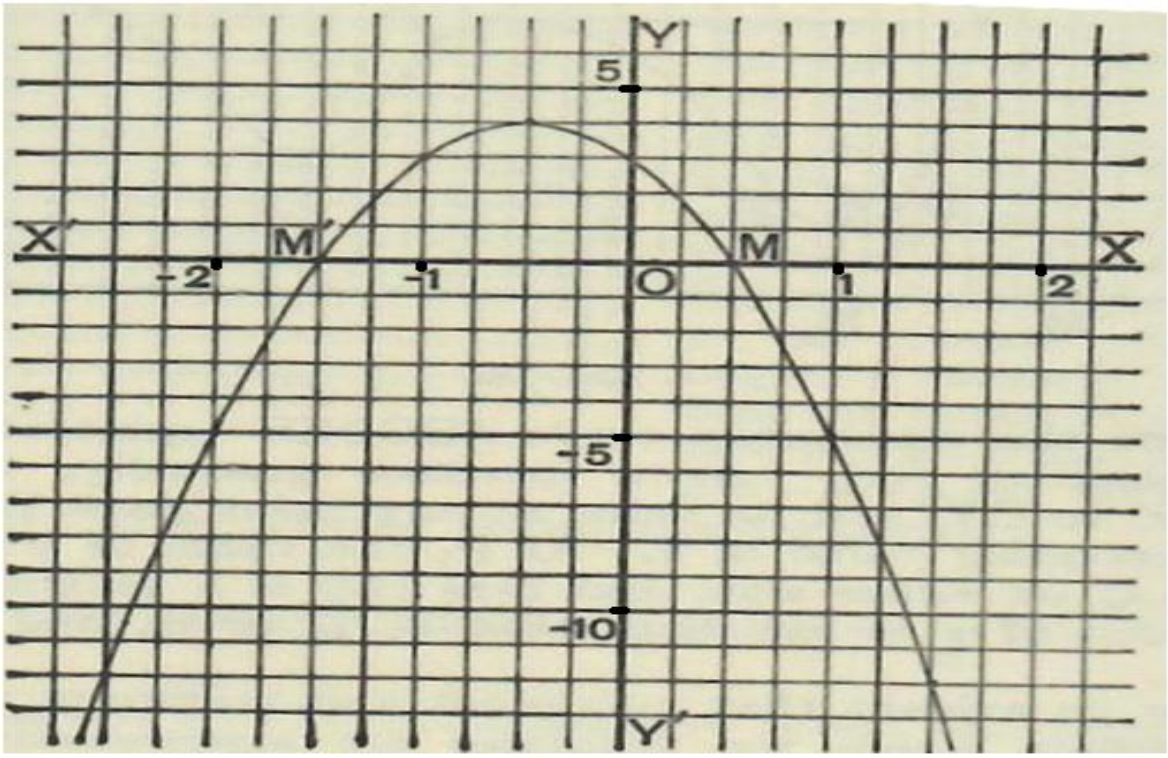
Write down three inequalities described by feasible **(6 marks)**

18.Figure 5 shows a quadratic graph crossing the $x - axis$ at points M and M̀.



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Find the equation of the curve in the form $y = ax^2 + bx + c$ **(6 marks)**

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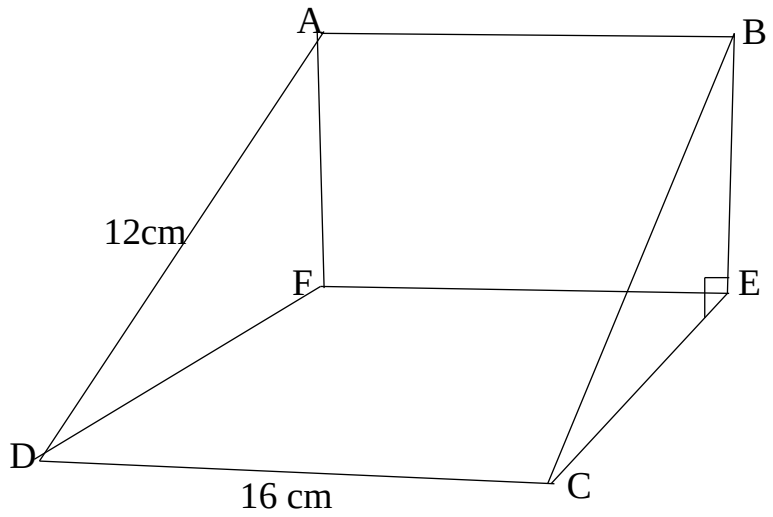
19. Figure 6 shows a triangular prism ABCDEF. AD= $12\ cm, DC = 16\ and\ angle\ BCE = 28°$



Calculate the total surface area of the prism. **(5 marks)**

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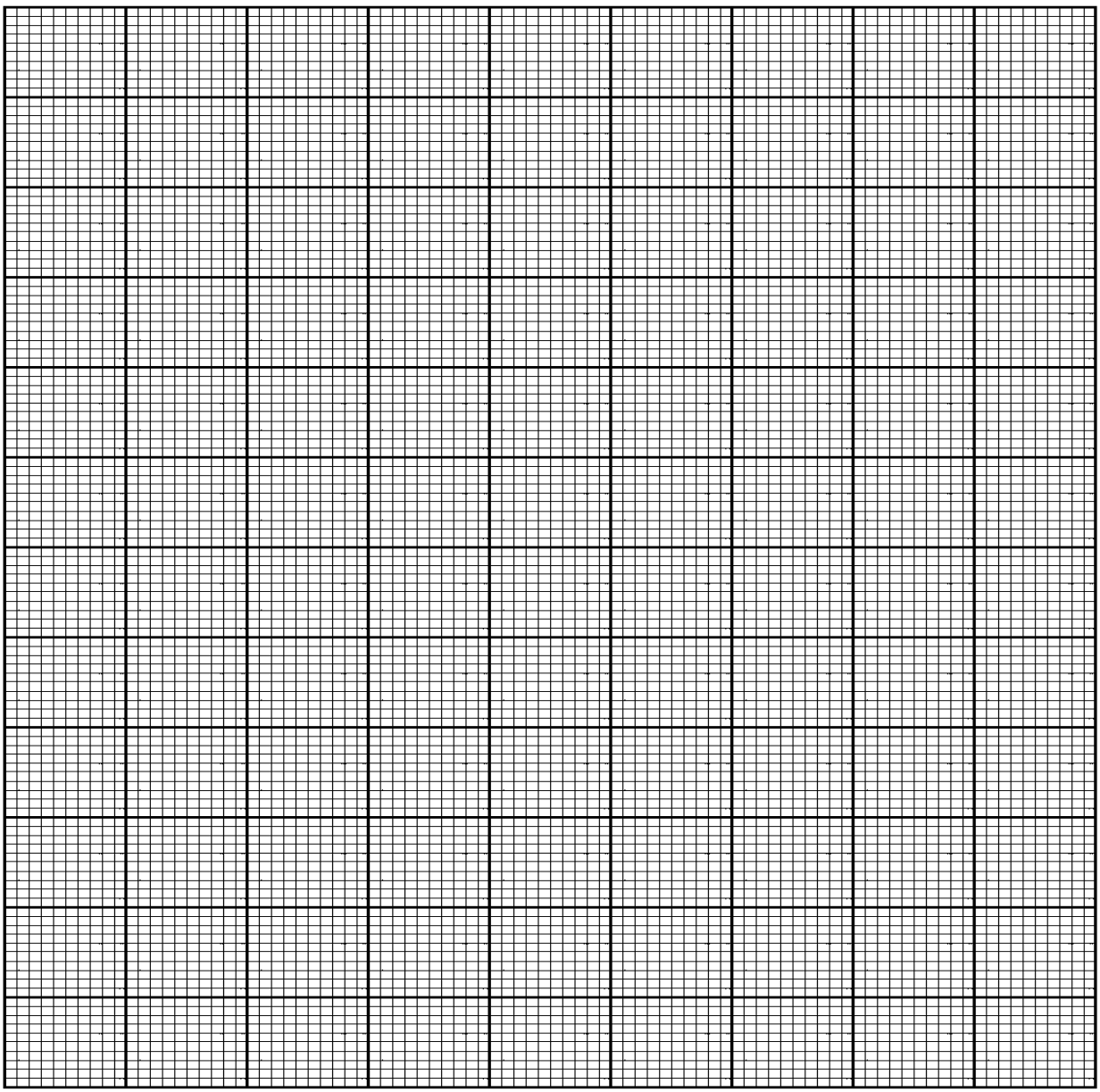
20. Using a scale of 2cm to represent 1 unit on both axes, draw a triangle whose vertices are $\mathbf{A}(0,0)$, $\mathbf{B}(5,1)$ and $\mathbf{C}(3,1)$.



On the same axis draw the image of triangle A, B, C under translation vector T= $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

(5 marks

END OF QUESTION PAPER